

ClimAlerts UI Components & Feature Explanations

Introduction

This document explains the purpose, meaning, and relevance of the UI components designed for the ClimAlerts system.

ClimAlerts is an intelligent climate-health monitoring platform focused on predicting disease outbreaks in the Volta Region using environmental and health data.

The UI components are designed to simulate a real-world public health monitoring system used by health authorities, analysts, and administrators.

1. Dashboard UI

Purpose

The Dashboard acts as the main control center of the system.

It gives administrators and health officials a quick overview of:

- current outbreak risks
 - environmental conditions
 - active alerts
 - AI recommendations
 - system activity
-

Dashboard Components

Overview Statistics Cards

These cards summarize important health and climate information.

Examples:

- Current temperature
- Humidity level
- Health risk level
- Active predictions
- Alerts sent today

Relevance

Allows users to instantly understand the current system condition without opening multiple pages.

Live Outbreak Activity

Shows recent disease predictions and outbreak activity.

Examples:

- Malaria risk in Ho Municipal
- Cholera warning in Keta South

Relevance

Helps authorities monitor outbreaks in real time.

AI Recommendations

Displays AI-generated advice based on predictions.

Example:

"High malaria outbreak risk detected in Ho Municipal. Recommended mosquito net distribution and drainage cleanup."

Relevance

Transforms the system from a simple monitoring tool into an intelligent decision-support platform.

Quick Actions

Provides shortcuts for important operations.

Examples:

- Trigger alert
- Upload dataset
- Retrain AI model

Relevance

Improves speed and efficiency during emergencies.

Recent Alerts

Displays recently sent SMS alerts.

Relevance

Allows administrators to confirm whether warning messages were delivered.

2. Predictions UI

Purpose

The Predictions page is the AI prediction engine interface.

It explains:

- what disease is predicted
 - confidence level
 - why the prediction happened
 - prediction trends over time
-

Predictions Components

Disease Forecasts

Shows predicted diseases.

Examples:

- Malaria risk
- Cholera outbreak risk
- Dengue fever risk

Relevance

Allows health officials to prepare before outbreaks become severe.

Confidence Score

Represents how confident the AI model is about the prediction.

Example:

- 87% confidence

Relevance

Helps users judge the reliability of predictions.

Higher confidence means stronger evidence from environmental and historical data.

Prediction Explanation

Explains why the prediction was made.

Example factors:

- rainfall increase
- humidity rise
- previous outbreak similarity

Relevance

Improves transparency and trust in the AI system.

This is called Explainable AI.

Prediction Trends

Displays how outbreak risk changes over time.

Usually shown using line graphs.

Relevance

Helps identify whether risk is increasing or decreasing.

Forecasting Timeline

Predicts outbreak conditions over future periods.

Examples:

- next 7 days

- next 14 days
- next 30 days

Relevance

Supports long-term planning and preparation.

3. Risk Map UI

Purpose

The Risk Map visually displays outbreak risk levels across districts in the Volta Region.

Risk Map Components

District Heatmap

Uses colors to represent outbreak severity.

Color Meaning:

- Green = Low Risk
- Yellow = Moderate Risk
- Orange = High Risk
- Red = Critical Risk

Relevance

Allows officials to quickly identify dangerous districts.

Volta Lake Overlay

Displays the Volta Lake on the map.

Relevance

Important because nearby water bodies influence mosquito breeding and disease spread.

District Details Panel

Shows detailed information for the selected district.

Examples:

- predicted cases
- rainfall
- humidity
- confidence score
- population at risk

Relevance

Provides localized outbreak intelligence.

Weather Overlays

Adds environmental data layers to the map.

Examples:

- rainfall intensity
- humidity zones
- temperature areas

Relevance

Shows environmental conditions contributing to outbreaks.

Outbreak Replay

A timeline playback system showing how outbreak risk changed over time.

Example:

- replaying risk progression from May 1 to May 30

Relevance

Helps analysts study outbreak evolution and patterns.

Useful for research and historical analysis.

4. Alerts & Messages UI

Purpose

Manages communication and emergency alert distribution.

Alerts Components

SMS Alert Composer

Used to create and send warning messages.

Relevance

Allows rapid communication with health officers and communities.

Recipient Groups

Different categories of recipients.

Examples:

- Health officers
- District coordinators
- Farmers
- General public

Relevance

Ensures alerts are delivered to the correct audience.

Delivery Status

Tracks SMS delivery progress.

Statuses:

- Sent
- Delivered
- Failed
- Pending

Relevance

Confirms whether emergency messages reached recipients.

Emergency Alert Templates

Predefined outbreak messages.

Relevance

Speeds up emergency communication.

Scheduled Alerts

Allows alerts to be sent automatically at specific times.

Relevance

Useful for recurring advisories and outbreak monitoring.

AI-Generated Alert Recommendation

Automatically generates outbreak advice.

Example:

"High malaria outbreak risk detected in Ho Municipal. Recommended immediate environmental sanitation."

Relevance

Adds intelligent automation to the alert system.

5. Data Source UI

Purpose

Handles data collection, synchronization, uploads, and API connections.

Data Source Components

CSV/Excel Upload

Allows manual upload of datasets.

Examples:

- weather records
- hospital case reports
- environmental data

Relevance

Supports data collection from offline or external sources.

API Integration

Connects the system to external services.

Examples:

- Ghana Meteorological Agency
- Ghana Health Service
- OpenWeather API

Relevance

Provides real-time data automatically.

API Sync

Automatically updates data from connected APIs.

Relevance

Ensures predictions use the latest environmental and health data.

Scheduled Sync

Automatically runs synchronization at specific intervals.

Examples:

- every hour
- every 6 hours
- daily

Relevance

Prevents outdated data from affecting predictions.

Data Validation

Checks uploaded data for:

- missing values
- duplicates
- invalid formats

Relevance

Improves prediction accuracy and system reliability.

Sync Logs

Displays synchronization history.

Relevance

Helps administrators track data updates and troubleshoot issues.

6. Users & Roles UI

Purpose

Controls system access and user permissions.

Users & Roles Components

User Management

Displays registered users.

Relevance

Allows administrators to manage staff access.

Roles & Permissions

Defines what users can access.

Examples:

- Admin
- Health Officer
- Data Analyst
- Viewer

Relevance

Protects sensitive system functions.

Activity Logs

Tracks user activities.

Examples:

- login time
- alerts sent
- data uploaded

Relevance

Improves accountability and monitoring.

7. Settings UI

Purpose

Controls system behavior, AI configuration, notifications, and backup management.

Settings Tabs Explained

SMS & API Settings

Purpose

Manages communication services and API connections.

Components

SMS Gateway Configuration

Connects the system to SMS providers.

Examples:

- Twilio
- Hubtel
- Africa's Talking

API Configuration

Stores API credentials for external data services.

Data Sync Settings

Controls automatic synchronization intervals.

Thresholds

Purpose

Controls how sensitive the outbreak prediction system is.

Disease Risk Thresholds

Sets risk detection levels for diseases.

Examples:

- malaria threshold
- cholera threshold
- dengue threshold

Relevance

Determines when the AI should classify a disease risk as dangerous.

Environmental Thresholds

Defines environmental trigger values.

Examples:

- rainfall level
- humidity percentage
- flood level
- temperature threshold

Relevance

Environmental conditions directly influence disease outbreaks.

Alert Severity Configuration

Defines risk ranges.

Examples:

- Low Risk
- Moderate Risk
- High Risk
- Critical Risk

Relevance

Controls how the system categorizes outbreak severity.

Live Risk Preview

Displays real-time risk calculations based on selected thresholds.

Relevance

Allows administrators to test and visualize threshold changes.

Backups

Purpose

Protects system data from loss.

Automatic Backup

Automatically saves system data.

Relevance

Prevents accidental data loss.

Backup Frequency

Defines how often backups occur.

Examples:

- hourly
- daily
- weekly

Relevance

Ensures data remains up to date.

Cloud Backup

Stores backups in cloud storage.

Relevance

Improves disaster recovery.

Restore Backup

Restores previous system data.

Relevance

Useful during system failure or corruption.

Backup Logs

Displays backup history.

Relevance

Helps administrators monitor backup health.

Notifications

Purpose

Controls system alerts and communication behavior.

Notification Preferences

Allows enabling/disabling:

- SMS alerts
- email alerts
- push notifications

Relevance

Customizes how users receive information.

Escalation Settings

Controls repeated alerts during emergencies.

Relevance

Ensures urgent outbreaks are not ignored.

AI Sensitivity

Purpose

Controls AI behavior and prediction strictness.

Detection Sensitivity

Determines how aggressively the AI detects outbreaks.

Examples:

- Low
- Balanced
- High
- Aggressive

Relevance

Higher sensitivity may detect outbreaks earlier but can increase false alarms.

Confidence Threshold

Minimum confidence required before showing predictions.

Relevance

Improves reliability of AI results.

Auto Retrain

Automatically retrains the AI model using new data.

Relevance

Keeps predictions accurate over time.

Explainable AI

Shows factors affecting predictions.

Relevance

Improves transparency and academic credibility.

System Preferences

Purpose

Controls general application behavior.

Theme Settings

Allows switching between light and dark mode.

Relevance

Improves user experience.

Language Settings

Changes application language.

Relevance

Supports localization.

Dashboard Preferences

Defines default startup pages and layout behavior.

Relevance

Improves workflow efficiency.

Profile & Account Settings UI

Purpose

Allows users to manage their personal account.

Components

Profile Information

Stores user details.

Examples:

- name
- email
- phone number
- role

Password Management

Allows changing passwords.

Relevance

Improves account security.

Notification Preferences

Controls personal alert preferences.

Relevance

Customizes user experience.

Session Activity

Displays login history and active sessions.

Relevance

Improves account monitoring.

Conclusion

The ClimAlerts UI system is designed to simulate a real-world climate-health intelligence platform.

Each component contributes to:

- outbreak prediction
- environmental monitoring
- emergency communication
- decision support
- health data management
- AI-assisted public health response

The design combines:

- modern dashboard principles
- public health monitoring concepts
- AI explainability
- enterprise SaaS architecture
- research-focused analytics

This makes ClimAlerts suitable for:

- academic projects
- public health research
- grant demonstrations
- outbreak preparedness systems
- smart health monitoring platforms